MUHAMMAD HARIS UMER

AI/ML Engineer | 2 Years | LLMs, RAG, Agents, FastAPI/Django
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PROFESSIONAL SUMMARY

AI/ML Engineer with 2 years of experience shipping LLM agents, RAG, and computer vision, plus Django, FastAPI, and Next.js services. At CoalDev, shipped Sortsy semantic search and an estimated 6,000-signal/sec Redis/Celery trading path.

PROFESSIONAL EXPERIENCE

AI/ML Engineer | CoalDev, Lahore, Pakistan *July 2025 – September 2026*

- Re-architected a real-time trading engine on Redis pub/sub and Celery to process ~6,000 market signals per second, raising throughput by an estimated 70–80%.
- Designed a multi-manager architecture that keeps live trading state in sync across accounts and strategies, and fixed PostgreSQL and Redis connection bottlenecks under peak load.
- Shipped Sortsy (sortsy.ai) features on a Django REST API, Celery workers, and a PySide6 desktop agent, with PostgreSQL, Redis, and WebSockets.
- Shipped a serverless vision pipeline on AWS Lambda and Amazon Bedrock that captions files and indexes them in Qdrant for semantic file search and retrieval.
- Deployed production n8n agent workflows with Anthropic Claude, Gemini embeddings and pgvector RAG.

AI/ML Intern | Nexpred Solutions, Lahore, Pakistan *April 2025 – June 2025*

- Developed multi-camera passenger queue analytics with YOLO, ByteTrack, and homography for queue length and density.
- Built an LLM intake and dispute assistant with Gemini tool calling for extraction, validation, and routing.
- Applied a reinforcement learning agent to aircraft parking and departure in a 3D hangar simulation.

AI/ML Intern | BitPix Soft, Lahore, Pakistan *July 2024 – September 2024*

- Engineered hierarchical chunking and agentic RAG for a legal-advisory assistant over long documents.
- Integrated speech-to-text and text-to-speech pipelines to power voice-enabled RAG chatbots.
- Generated synthetic training data with LLMs to widen chatbot response coverage.

EDUCATION

Bachelor of Science in Computer Science | FAST-NUCES, Lahore, Pakistan *August 2021 – June 2025*

PROJECTS

Aircraft Parts Quote Automation | LLM Client Project *September 2025 – October 2025*

- Built an LLM pipeline that reads supplier quote threads from Gmail into Pydantic-validated structured outputs.
- Ranked suppliers per part and wrote the shortlist to Google Sheets, replacing manual inbox review.
- Supported OpenAI and Anthropic behind the same extractor, with secrets kept in environment variables.

Artikon — Agentic Comic Generator | FAST-NUCES Group Final Year Project *August 2024 – June 2025*

- Built a script-to-comic pipeline that holds character identity across panels, served in a Next.js app.
- Orchestrated LangGraph stages for character resolve, mask, generate, composite, and an identity check.
- Conditioned Stable Diffusion with ControlNet and IP-Adapter, and ranked LoRA style tags by embedding similarity.

Visible Electrical Defect Detection | Computer Vision *August 2026 – September 2026*

- Trained YOLOv8n detectors on 7,000+ labeled images to find open, short, and damage defects in PCBs and cables.
- Scored 0.888 mAP50 on outdoor cable damage and 0.831 mAP50 on a 4-class PCB and cable model.
- Raised precision from 0.81 to 0.86 with CLAHE and sharpening on 484 test images. A screening aid, not a safety certificate.

TECHNICAL SKILLS

LLMs & Agents: LLMs, RAG, AI Agents, Tool Calling, Prompt Engineering, LangGraph
Models & APIs: OpenAI, Anthropic Claude, Google Gemini, Amazon Bedrock, Hugging Face, Stable Diffusion
ML & Vision: PyTorch, Transformers, YOLO, ByteTrack, OpenCV, ControlNet, IP-Adapter, Reinforcement Learning
Vector & Data: Qdrant, pgvector, Embeddings, Semantic Search, PostgreSQL, Redis, SQL
Backend: Django REST Framework, FastAPI, Flask, Celery, REST APIs, WebSockets, Event-Driven Architecture
Languages & Tools: Python, TypeScript, JavaScript, Next.js, AWS Lambda, Git